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Lesson Notes

MODULE 1 | LESSON 3

FROM UTILITY THEORY TO CLASSICAL PORTFOLIO THEORY

Reading Time	120 minutes
Prior Knowledge	Matrices, algebra, some differential calculus
Keywords	Utility theory and relevant specific examples (CARA, CRRA, logarithmic utility, quadratic utility), The problem of maximizing utility and Markowitz portfolio theory, Quadratic programming

In the previous lesson, we looked at portfolio risk and how to measure the severity of possible losses using several approaches to VaR. In this lesson, we are looking at how to build optimal portfolios and how to incorporate certain constraints.

1. Elements of Utility Theory

The mean-variance model of asset choice has been used extensively in finance since Markowitz developed it back in 1952. To understand some of the nuances behind this model, one has to start with at least a modicum of utility theory.

There is clearly a difference between deciding which used car to buy, what brand of hamburger to eat, which project to choose for a company, or, generally, choosing among outcomes we do not know in advance. If a person decides to buy a share of Apple or Bitcoin, they do not know the outcome of their choice because the stock price follows a stochastic process. If one buys a meal, then we can assume that one knows.

There is also more than one way of not knowing the outcomes of certain events. For example, if we decide to play a roulette wheel, we do not know the outcome, but we could compute the probabilities for every outcome. However, we would not be able to do that in the case of betting on horse races. This has a consequence: Everyone who can perform the required calculations would agree on the likelihood of each outcome while playing a roulette wheel, but this is not so when betting on horse races. When the probabilities of outcomes are objective (i.e., different people w compute the same probability for an event), we have what we could describe as **risk**; on the

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Traditional portfolio theory assumes that all individuals are rational and capable of performing the required calculations to optimize their behavior and choices. Is that reasonable? Perhaps not, but it allows us to use higher mathematics to solve problems. Non-traditional approaches to portfolio theory will remove the assumption that all individuals have perfect information and/or optimize their every single decision. But that is a story for another part of the course.

2. Expected Values do not Work Well

When dealing with two or more risky situations, most of us, acting rationally, would find it sensible to choose the action that guarantees the highest average payoff, whenever the expected payoffs are greater than the cost of betting, of course. If it were not the case, then we would decide not to take the bet(s).

This is what one learns from early on dealing with the tools of probability and statistics. This approach is known as the **expected value approach**. However, this type of strategy can produce unforeseen consequences.

For instance, there is a famous example attributed to Nicolaus Bernoulli in 1713 and known as the **St. Petersburg paradox** , which we will cover in a moment, and a second one which shows some of the complications that ensue if we use an expected value approach, as well.

Example 1: (St. Petersburg paradox). Given a fair coin, we play the following game:

- **Step 1**. Flip a coin. If the outcome is heads (H), we win $\$2$;
- **Step 1**. If the outcome is tails (T), we flip the coin again.
- **Step 2**. If the coin comes up H, we win $\$2^2$;
- **Step 2**. If the coin comes up T, we flip the coin again: $\ldots$
- **Step n**. If the coin comes up H, we win $\$2^n$;
- **Step n**. If the coin comes up T, we flip the coin again: $\ldots$ forever.

The expected value for this game is then relatively easy to compute:

$$\text{expected value} = \sum_{n=1}^{\infty} 2^n \cdot \frac{1}{2^n} = \sum_{n=1}^{\infty} 1 = \infty.$$

How many times should we pay to play this game? Apparently, an infinite amount, i.e., one should play the game at any price, if offered the opportunity.

However, Daniel Bernoulli in his *Specimen Theoriae Novae de Mensura Sortis* (Exposition of a New Theory on the Measurement of Risk) in 1738 published in the *Commentarii Academiae Scientiarum Imperialis Petropolitanae* (Papers of the Imperial Academy of Sciences in Petersburg) reported that his cousin Nicolaus Bernoulli wrote in a letter to him that—after deriving this result and describing the game with an initial stake of 1 ducat—although the value of the expectation was infinitely great, "it [illegible] to be admitted that any fairly reasonable man would sell his chance, with great pleasure, for t[illegible] ducats."

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of the Econometric Society, pp. 23–36, 1954.

So, it seems that although one cannot lose at this game, 20 ducats may induce a person to choose this option, despite the lesser payoff. The second example is equally dramatic.

Example 2. (The pauper and the lottery). A poor person receives a magic lottery ticket, one that has a 50% chance to win \$1 million and a 50% chance to win \$0. A rich person offers to buy the ticket off the poor person for \$499,999 with certainty.

If the expected value approach held, the poor person would refuse the offer. However, most of us would agree that this would be unlikely as the difference between the expected value, \$500,000, and \$499,999 is minuscule, while the difference between \$0 and \$499,999 is massive for a poor person. The difference between \$1 million and \$499,999 is large but not as dramatic. However, choosing the \$499,999 invalidates the expected value approach, and yet that is what most people would do as shown in experiments.

The St. Petersburg paradox is so well-known that it has its own entries in encyclopedias and it can be found in discussions ranging from philosophy to probability to finance, etc. (for instance, Stanford Encyclopedia of Philosophy).

This paradox has faced several criticisms among which is the fact that "anyone who offers to let the agent play the St. Petersburg gamble is a liar, for he is pretending to have an indefinitely large bank" (Jeffrey), but it has also found those who defend it. Nevertheless, the second example is much less debatable and shows that the attitude towards wealth cannot be regarded just as an attitude towards an amount of money or monetary means and that we will have to consider it in terms of its utility to an economic agent.

To this purpose, casinos in Las Vegas fix a maximum bet of \$5,000 on roulette. Hence, if someone lost 12 times in a row, they would be unable to bet $\$2^{13} = \$8,192$. Consequently, the St. Petersburg gamble could not happen in Las Vegas.

3. Expected Utility Theory

Daniel Bernoulli suggested that people maximize expected utility, not expected value. To carry forward this program one has to introduce a **utility function**, say $U(w)$, and which represents the expected utility of an amount of wealth w. The function should be increasing ("more is better"), but the marginal utility should be decreasing (the value of an additional dollar value gets lower with each additional amount of wealth). Mathematically, this means that a utility function, $U(\cdot)$, should have the following properties:

$$dU(w)/dw > 0, \quad \text{and} \quad d^2U(w)/dw^2 < 0.$$

In particular, the negative second derivative alerts us to the fact that $U(\cdot)$ should be concave.

Example 3. (St. Petersburg paradox, continued). Let's assume that the player has the following utility function: $U(w) = \ln(w)$. It is easy to check that this utility function satisfies the two requirements we asked of a utility function. Then:

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The implications are even further reaching. In fact, we could consider the fact that if a person has an initial wealth of w_0,, then the value of the expected utility would be

$$E[U(w)] = \sum_{n=1}^{\infty} ln(2^n + w_0)\frac{1}{2^n} < \infty,$$

and it would be different, depending on how wealthy the gambler is originally. Thus, different agents would have different valuations for the same gambling games.

Unlike in the case of the expected value approach, Bernoulli is proposing that the majority of people do not extract utility from wealth per se, but rather, they extract utility from consuming goods, which is achieved by using that wealth to purchase the goods. This means that there is possibly a complicated nonlinear relationship between wealth and utility of wealth (that is, consumption). Nevertheless, Bernoulli tells us that this relationship is characterized by a utility function that is both increasing in wealth and concave (that is, it is increasing, but at a decreasing rate). Once we agree to adopt a utility theory approach, then we would choose x over y if $E[U(x)] > E[U(y)]$. We are still using the expected value, but in a very different fashion.

At this time, we assume that individuals are alive for only one period, which implies that individuals make decisions to use all their final wealth immediately to purchase/consume goods and services.

4. Risk-Averse Agents and Concave Utility Functions

We need to start with a formal definition, one that allows us to avoid running into troubles when using the everyday language instead.

Definition. An agent is **risk-averse** if, at any wealth level w, they dislike every lottery with an expected payoff of zero; i.e.,

$$E[U(w + z)] \leq E[U(w)]$$

for every lottery z such that $E[z] = 0$.

We should notice that every lottery, z, is a random variable and, as such, it can be written as $z = E[z] + (z - E[z])$; i.e., z can be decomposed into its expected payoff $E[z]$ and a zero mean lottery $z - E[z]$. Then, the definition of a risk-averse agent as given above can be reformulated by saying that:

A risk-averse agent always prefers receiving the expected outcome of a lottery with probability 1, rather than the lottery itself. For an expected utility maximizer, this implies that for any lottery z and for any initial wealth w :

$$E[U(w + z)] \leq U(w + E[z]).$$

The last relationship is always true for a concave function $U(\cdot)$ (Jensen's inequality for random variables), and therefore, we can infer that assuming a concave utility function is equivalent to saying that the economic agent is risk-averse. In fact, we have the following stronger result:

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Consider now a risk-averse agent, i.e., someone who dislikes zero-mean risk. That does not mean that they avoid any lottery like the plague. Instead, the agent may still like risky lotteries if the expected payoff is sufficiently large. More specifically, risk-averse investors may still want to buy risky assets if the expected return exceeds the cost to buy insurance.

This raises the following question: How is the optimal trade-off between the expected gain and the degrees of risk determined? Individuals are different: Some are willing to spend more than others to get rid of a specific risk. If π represents the risk premium, then for an agent with utility function $U(\cdot)$ and initial wealth w, the risk premium should satisfy the following equation:

$$E[U(w+z)] = U(w-\pi).$$

The agent ends up with the same amount of utility either by accepting the risk of the zero-mean lottery z or by paying the risk premium π to avoid the uncertainty of the lottery.

Since we know from earlier that, for a risk-averse agent, we have $E[U(w+z) \leq U(w+E[z])$, π must be nonnegative! The risk-premium is a complex function of the probability distribution of z and of the utility function $U(\cdot)$. Nevertheless, if we consider infinitesimal changes in the amounts of risk z, we could use a second-order Taylor approximation:

$$U(w-\pi) \approx U(w) - U'(w)\pi$$

and

$$\begin{aligned} E[U(w+z)] &\approx E[U(w) + U'(w)z + \frac{1}{2}U''(w)z^2] \\ &= U(w) + U'(w)E[z] + \frac{1}{2}U''(w)E[z^2] \\ &= U(w) + \frac{1}{2}U''(w)\sigma_z^2 \end{aligned}$$

where we used the fact that $E[z] = 0$ and let $\sigma_z^2 := E[z^2]$, the variance of the zero-mean lottery. Then, replacing what we just found in the earlier equation $E[U(w+z)] = U(w-\pi)$, we find that

$$\pi \approx \frac{1}{2}\sigma^2 \frac{-U''(w)}{U'(w)}.$$

Note that π is nonnegative since $U''(w) < 0$ and $U'(w) > 0$ because of the characteristics of $U(\cdot)$ for risk-averse investors. Incidentally, the function

$$A(w) := \frac{-U''(w)}{U'(w)}$$

which is called the **Arrow-Pratt measure of absolute risk-aversion** provides a good measure of risk aversion. The higher $A(w)$, the higher the risk aversion of the agent. We can see that this is the case because, everything else being equal, it tells us what amount an agent is willing to pay for insurance. The more risk-averse they are, the more they would pay to eliminate the risk.

A TOP r measure frequently used is the following:

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which is called a **measure of relative risk-aversion**. To understand the reason behind this name, one should notice that

$$\pi_r = \frac{\pi}{w} \approx -\frac{1}{2}\frac{U''(w)w}{U'(w)}\frac{\sigma_z^2}{w^2} = -\frac{1}{2}\frac{U''(w)w]}{U'(w)}\sigma_{z/w}^2 = \frac{1}{2}R(w)\sigma_{z/w}^2.$$

In the case of relative risk aversion, it is the variance of z/w and not the variance of just z that is relevant, hence the name of relative risk aversion.

5. Some Relevant Utility Functions

So far, our treatment has been very theoretical and to a large extent expected utility theory is. However, to become more acquainted with the concepts and also to gain some ability to read technical papers and literature, it is convenient to look at a few "practical" examples and learn what to look for.

Example 4. (CARA utility function). Suppose that the utility function is given by

$$U(w) = 1 - e^{-\alpha w}, \;\; \alpha > 0.$$

Then $U'(w) = \alpha e^{-\alpha w}$ and $U''(w) = -\alpha^2 e^{-\alpha w}$. Therefore

$$A(w) = \frac{-U''(w)}{U'(w)} = \alpha.$$

i.e., this utility function has a **constant absolute risk aversion**, and for that reason, it is known as a CARA utility function. To check on an important property of this utility function, let's assume that a certain payoff of d is preferred to a lottery z. This means that $U(d) \geq E[U(z)] = \sum_z U(z)Pr[z]$. Since the utility function is the CARA utility function, this implies that

$$1 - e^{-\alpha d} \geq \sum_z (1 - e^{\alpha z})Pr[z] \quad \text{or}$$

$$1 - e^{-\alpha d} \geq 1 - \sum_z e^{-\alpha z}Pr[z] \quad \text{or}$$

$$e^{-\alpha d} \leq \sum_z e^{-\alpha z}Pr[z].$$

Consider now another amount $d' > 0$. Then, from the last inequality, we find

$$e^{-\alpha d'}e^{-\alpha d} \leq e^{-\alpha d'}\sum_z e^{-\alpha z}Pr[z],$$

$$1 - e^{-\alpha d'}e^{-\alpha d} \geq 1 - e^{-\alpha d'}\sum_z e^{-\alpha z}Pr[z].$$

The last inequality is easily seen to be

$$U(d + d') \geq \sum_z U(z + d')Pr[z] = E[U(z + d')],$$

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or losing $75 is the same regardless of whether the initial wealth is, say $100, $100,000, or $ 1,000,000. Is this realistic? Probably not.

Example 5. (CRRA utility function.)

Let

$$U(w) = \frac{w^{1-\beta} - 1}{1-\beta}, \ \ \beta > 0.$$

In this case, one shows with a little bit of calculus and algebra that

$$R(w) := wA(w) = \beta,$$

i.e., the relative risk aversion is constant, or in plain English, proportional changes in prizes for the lottery do not affect the attitude towards risk. For this very reason, this utility function is known as CRRA (**constant relative risk aversion)**.

Example 6. (Logarithmic utility function). The log-utility function

$$U(w) = ln(w)$$

has the property that

$$A(w) = 1/w,$$

i.e., absolute risk aversion declines with the level of wealth. Instead, relative risk aversion is constant.

Example 7 (Quadratic utility function).
For a quadratic utility function, i.e.

$$U(w) = w - \gamma w^2, \ \ \gamma > 0,$$

(what happens if we used $+\gamma$ instead of $-\gamma$?) One finds that

$$A(w) = \frac{2\gamma}{1 - 2\gamma w}$$

and $A'(w) = \frac{4\gamma^2}{(1-2\gamma w)^2} > 0$, which implies that the absolute risk aversion is increasing with w, which makes the quadratic function suitable to model investors who reduce the nominal amount invested in risky assets as their wealth increases. This makes risky investments inferior goods, unfortunately!

6. Types of Investors and Their Utility Functions

Because this course is aimed at students of finance, it is worthwhile to conclude this section by looking at how to categorize the different types of agents.

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or taking their chances. On the other hand, a {\bf risk-seeking agent} is a decision-maker who prefers a risky payoff over a sure amount equal to its expected value.

All three types of agents have a utility function that is increasing, $U'(\cdot) > 0$. However, unlike in the case of the risk-averse agent, for the risk-neutral investors, it is $U''(\cdot) = 0$, i.e., the utility function is linear. On the other hand, for risk-seeking investors, $U''(\cdot) > 0$ or the utility function is convex.

If we look at the equation for π, we could also categorize the different types of agents by stating that risk-averse agents are willing to pay to eliminate risk while risk-seeking agents are willing to pay to take risk.

Traditional portfolio theory assumes that agents are risk-averse, but behavioral finance does not. And, in real life, we come across individuals who derive utility from participating in "lotteries," whatever lottery might mean to each of them.

There is another way to look at the three different types of economic agents. How do they behave when they need to invest?

- Risk-averse investors believe that "something is better than nothing" and like to play it safe. They tend to choose safer investments to place their assets. Of course, their payout still needs to meet the investors' demand for a return on investment.
- Risk-neutral investors simply select the investment with the highest expected return and do not take into account the risk of investments in their decision-making process.
- Finally, risk-seeking investors tend to invest in stocks with high betas, speculative investments, junk bonds. Sometimes, they are outright gamblers. They are willing to take on a considerable amount of risk as long as the potential return is worth the risk of their investment.

7. From Utility Theory to Classical Portfolio Theory

The mean variance model of asset choice introduced by H. Markowitz in 1952 has been already seen by students in previous classes. Essentially, we are told that, for instance, to find the minimum variance portfolio compatible with a given desired return, one should solve the following problem:

$$\begin{aligned} &\min_{w_1,\ldots,w_m} \sum_{i=1}^{m}\sum_{j=1}^{m} \sigma_{ij} w_i w_j \\ &\text{subject to } \sum_{i=1}^{m} w_i r_i = r_{min} \\ &\sum_{i=1}^{m} w_i = 1. \end{aligned}$$

We can generalize this problem by writing it in a different form that is more suitable for the developments that we are going to pursue in the next lectures. To this purpose, we write it as follows:

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$$\text{subject to } \sum_{i=1}^{m} w_i = 1.$$

In other words, instead of just finding the portfolio that has the lowest variance among those who yields a given return r_p, we just find the portfolio that maximizes the following expected utility function:

$$E[U(W)] = E[W] - \frac{\lambda}{2}\text{Var}[W]$$

where W denotes wealth. In the case of a portfolio investment $W = \sum_{i=1}^{m} w_i A_i$ where the A_i's are the assets that form the portfolio and the w_i's are the weights of those assets in the portfolio, respectively. We should notice that the optimization process will try to select weights that maximize return
and, at the same time, minimize variance. The presence of the $\frac{\lambda}{2}$ factor allows us to model different levels of risk aversion (the higher λ the more risk-averse the investor).

We have made a big deal about anchoring portfolio decisions to utility theory. So, it is legitimate to ask ourselves how we can do that. To this purpose, let's consider an investor who has an exponential utility; i.e.:

$$U(W) = -e^{-\lambda W}$$

where $\lambda > 0$ and W as usual denotes wealth. It is easy to check that $A(W) = \lambda$, i.e. the absolute risk aversion is constant and the larger
λ is, the more risk averse the agent is, which matches what we have in our optimization program. Now, we make a further assumption and assume that the wealth, W, is normally distributed with mean μ variance σ^2. In other words, we are assuming that the probability density function for W is

$$f(W) = \frac{1}{\sigma\sqrt{2\pi}} e^{-(1/2\sigma^2)(W-\mu)^2}.$$

Then, the expected utility is given by

$$E[U(W)] = \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{\infty} -e^{-\lambda w} e^{-(1/2\sigma^2)(W-\mu)^2}\, dW$$

or,

$$E[U(W)] = \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{\infty} -e^{-(\lambda W + (1/2\sigma^2)(W-\mu)^2)}\, dW.$$

Now, a simple algebraic trick which is common among Statisticians and Probabilists when using normal random variables the way we are doing here
will prove very useful. More specifically, the exponent in the exponential function can be written in form:

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Using this equality in the previous formula, we get

$$E[U(W)] = \frac{e^{-\lambda(\mu-(1/2)\lambda\sigma^2)}}{\sigma\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{(W-\mu+\lambda\sigma^2)^2}{2\sigma^2}} dW.$$

At this point we note that the exponential inside the integral sign is just the pdf of a $N(\mu - \lambda\sigma^2, \sigma^2)$ random variable and thus

$$\frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{(W-\mu+\lambda\sigma^2)^2}{2\sigma^2}} dW = 1.$$

Therefore we have just shown that

$$E[U(w)] = -e^{-\lambda(\mu-(1/2)\lambda\sigma^2)}$$

which does not even depend directly on W. Finally, if we want to maximize $E[U(w)]$, it is clear that we need maximize the expression

$$\mu - \frac{\lambda}{2}\sigma^2.$$

Considering that the mean for our portfolio is given by $\sum_{i=1}^{m} w_i r_i$ and the variance by $\sum_{i=1}^{m}\sum_{j=1}^{m} \sigma_{ij} w_i w_j$, we have shown
that the problem is indeed anchored in utility theory, at least as long as we use an exponential utility function as done above.

Remark 1. Someone may wonder whether if we could have found the same result using a quadratic utility function like $U(W) = \alpha W + \beta W^2$ and without making any assumption about the distribution of W. Well, not quite. In this case, we would find that

$$E[U(w)] = \alpha E[W] + \beta E[W^2] = \alpha E[W] + \beta[\mathbf{Var}[W] + (E[W])^2)$$

and we notice that we have an extra term, $(E[W])^2$, inside the square brackets on the right-hand side. However if, say $E[W] = 0.10$, then $(E[W])^2 = 0.01$ and therefore we could considered it negligible, but technically we do not get the same problem as written at the beginning of this section. Of course, it would be an equally easy problem to solve replacing the variance of the wealth with its second moment instead.

Remark 2. The result was obtained by assuming a specific utility function for our generic investor. What happens if we are not willing to assume that particular utility function? To investigate this point in a very general setup, let's assume that we have a utility function $U(W)$ whose functional form is arbitrary. Take a Taylor's series expansion of this function about $E[w]$. But, do this in this way:

$$U(W) = U(E[W)] + (W - E[W]))$$

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where

$$R = \sum_{k=3}^{\infty} \frac{1}{k!} U^{(k)}(E[W])(W - E[W])^k.$$

In Markowitz setup, there is an assumption that investors decide about their investments based on the first two moments, i.e. $R = 0.$ Then, take expectation on both sides and we have

$$E[U(W)] = U(E[W]) + \frac{1}{2} U''(E[W]) Var[W].$$

If we assume that the typical investor is risk averse, then $U''(E[W]) < 0.$ This leads us to the classical MVO setup, but only if we are willing to assume that the extra terms (moments) in the Taylor's expansion can be set to zero or are at least negligible. Whether this is justified or not will depend on the true and usually unknown distribution of the wealth process $W.$ Notice for instance that if assumed again that W is normally distributed, then all odd central moments would be zero, but not the even moments. However, we could show that the even moments are related to the variance, but we will not pursue this here as it gets complicated and fast. Or we would need to assume a utility function whose derivatives past the second are null. The quadratic utility function seen above has this feature, but it does not cut the profile completely as we saw already.

Research has shown that if there is a riskless asset, then the distribution of every portfolio is determined by its mean and variance if and only if the random returns are a linear transformation of a spherically distributed random vector. This includes, among other distribution, the normal distribution. If there are no riskless assets, but returns are such that the last $n - 1$ components are spherically distributed conditional on the first component, which is allowed to have any distribution, then the distribution is also determined by its mean and variance. Students interested about this parts could read the following paper: G. Chamberlain, *A Characterization of the Distributions That Imply Mean-Variance Utility Functions*, Journal of Economic Theory, 29, p. 185-201 (1983) which contains the result mentioned above and a relatively rich list of references that discuss the issue going back to the 1960s.

8. Conclusion

In this lesson, we showed that economic agents are not driven simply by monetary amounts. The same monetary amount means different things to different people, depending on their initial wealth. This created the need to introduced utility theory to describe the actions and decisions of economic agents.

After introducing utility theory, we applied it to the classical portfolio optimization problem and showed that, in its standard formulation, such problem is oversimplified since quadratic utility functions are not completely satisfactory, but it has the merit to make the problem relatively simple to deal with in a mathematical rigorous manner.

In the next lesson, we will study how to solve the problem of finding optimal portfolios, determining t cient frontiers, identifying minimum variance and the tangency portfolios, and handling some

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